

Alannah “Al” Oleson, Ph.D.

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I envision a world where *all* computer science students are equipped with the skills they need to make accessible, inclusive, and ethical technology, empowering them to create more just technological futures. To achieve this, I work alongside post-secondary educators to develop and evaluate pedagogical interventions for critical HCI and CS education through qualitative, mixed-method, and participatory action research investigations.

Education

- 2018-2023 **University of Washington**, Seattle, WA, USA
Ph.D. in Information Science
Dissertation: *Integrating Inclusive Design and Computing Education*
Advisor: Amy J. Ko
- 2021 **University of Washington**, Seattle, WA, USA
M.S. in Information Science
General Exam: *Toward Justice-Centered Software Design & Development*
Advisor: Amy. J. Ko
- 2014-2018 **Oregon State University**, Corvallis, OR, USA
Honors B.S. in Computer Science, summa cum laude
Thesis: *Pedagogical Content Knowledge for Teaching Inclusive Software Design*
Advisor: Margaret M. Burnett

Research Experience

- Sept. 2024 **Assistant Professor**
-present Ritchie School of Engineering and Computer Science, University of Denver
Director of TEACUP (Teaching Empathy, Agency, and Criticality for Unlearning Power) Lab
Exploring novel integrations of critical computing ideas into CS education through several projects, including collaborations around CS students' ethical agency in technology design, critical AI literacies for humanities education, and broadening participation of low-income students in computing education.
- 2023-2024 **Postdoctoral Scholar & Co-Founder**
Center for Learning, Computing, and Imagination, University of Washington
Led an Action Research study with a community of computing educators on best practices for teaching and learning critical interface design skills within postsecondary computing courses.
- 2018-2023 **Graduate Research Assistant**
Code & Cognition Lab, University of Washington (Advisor: Amy J. Ko)
Led several research projects on justice-centered computing education and inclusive technology design education, including (though not limited to) a critical review of demographic data trends within computing education research and a freely-accessible digital book to help computing faculty integrate accessibility topics into their college-level computer science courses.
- 2017-2018 **Procedural Imaging Group Intern**
Adobe Research (Mentors: Cynthia Lu, José Echevarria, Radomir Mech)
Designed, conducted formative user research for, implemented, and patented a proof-of-concept augmented reality (AR) guided selfie and portrait-taking interface for mobile photography.

2014-2018 **Undergraduate Research Assistant**

EUSES / Gender HCI Lab, Oregon State University (Advisor: Margaret M. Burnett)

Supported and later led projects on lowering barriers to end-user software engineering through just-in-time help systems and gender-inclusive interface design methods for industry practitioners.

Funding & Grants

2026 (under review, recommended for consideration) **Collaborative Research: IUOE: ESL Level 2: People First: Embedding Socially Responsible Design Across the CS Curriculum**

National Science Foundation

*Lead: Patricia Morreale (Kean); Institutional PIs: **Alannah Oleson** (U. Denver), Margaret Burnett, Christopher Hundhausen (Oregon State), Briana Morrison (UVA)*

*NSF IUOE 23-510: Improving Undergraduate STEM Education, Level 2 (Institutional Transformation)
\$400,942 over 3 years (\$165,951 for U. Denver)*

Leveraging successful teacher education workshop structures, we will work alongside CS faculty to integrate socially responsible design ideas across each institution's 4-year core curriculum and evaluate impacts on students' learning outcomes, sense of belonging, and perceptions of education climate.

CHAI: Career-Ready Humanities-based AI Education Modules

Mozilla Foundation: Mozilla Responsible Computing Challenge

*PI: **Alannah Oleson**; Co-PI: Benjamin Xie (U. Denver)*

Faculty Partners: Lindsey Feitz (Gender & Women's Studies), Kaitlyn Sims (Public Policy), Laurel Taylor (Japanese Studies)

\$125,000 over 2 years

Artificial Intelligence (AI) is transforming necessary workplace proficiencies, but most work exploring these focuses on STEM education. Together with faculty partners from humanities & social science (HSS) disciplines, we propose a set of responsible AI education modules (curricular integrations, seminars, and community-based projects) to help HSS students develop AI literacies that are relevant to their disciplines.

RELATE Scholars: Relationships Elevating Students to Achieve Transformational Educations

National Science Foundation Award # 2424972

*PI: Michelle Sabick; Co-PI: Scott Leutenegger; Co-Is: **Alannah Oleson**; Benjamin Xie (U. Denver)*

NSF 25-514: NSF Scholarships in Science, Technology, Engineering, and Mathematics Program (S-STEM) Track 1 (Institutional Capacity Building)

\$996,598 over 5 years

Building upon relationship-rich education principles, we will support the retention and graduation of high-achieving, low-income computer science students with demonstrated financial need through a cohort-based mentoring model and financial assistance, exploring which of these factors most impacts students' success.

Measuring Semantic Fidelity in LLM-Based Privacy Policy Interpretation for Older Adults

Center for Ethical Generative Artificial Intelligence Applications, U. Denver

*PI: **Alannah Oleson**; Co-Is: Iffat Anjum, Suleiman Saka (U. Denver)*

*NSF IUOE 23-510: Improving Undergraduate STEM Education, Level 2 (Institutional Transformation)
\$7,000 over 1 year*

Based in human-centered design and accessibility principles, we propose a novel system which leverages LLM-based simplification privacy policies for healthcare IoT (Internet of Things) technologies to support older adults' (65+) agentic decision-making around the collection and use of their healthcare data.

Exploring Accessible Teaching Practices in Computer Science Syllabi, Univ. of Denver

*Mentor: **Alannah Oleson**; Student: Muskan Fatima*

Discovery PINS Student Signature Work Grant (advisor)

\$3,800 for Winter-Summer 2026 project work

Visiting Researcher Grant, Uppsala University

Equal Opportunities Group, Faculty of Science & Technology

To establish international collaborations and develop ongoing work on ethical agency in CS education. 50,000kr (~\$5,250) to support travel and lodging in August-September 2026.

2025 Computing Students' Perceptions of their Own Ethical Agency in Technology Design Processes

Research Innovation Scotland (RIS) International Collaboration Fund

*PIs: Matthew Barr, Maria Kallia (U. Glasgow), **Alannah Oleson** (U. Denver), Bedour Alshaigy (Uppsala U.)*

£4,520 (~\$5,940) to support initial collaboration, travel, and pilot studies in 2026

Computer science (CS) students often believe considering ethics, accessibility, and inclusivity is "not their job" - someone else will handle these concerns. We will develop and validate a survey instrument measuring ethical awareness and agency, building on established "computing capital" frameworks. Findings will reveal how university programs shape - or fail to shape - ethical awareness, providing the basis for a future longitudinal, multi-national research project on ethical agency in computing education.

Evaluating a Pedagogical Resource for Teaching Accessible Computing, Univ. of Denver

*Mentor: **Alannah Oleson**; Student: Muskan Fatima*

Discovery PINS Student Signature Work Grant (advisor)

\$1,500 for Spring 2025 project work

Visiting Researcher Grant, Uppsala University

Equal Opportunities Group, Faculty of Science & Technology

To promote visibility of CS ed. research in the department and establish international collaborations.

50,000kr (~\$5,250) to support travel and lodging in June 2025

KEEN National Conference Travel Grant, Univ. of Denver

Covers registration, lodging, and airfare (~\$2000) to attend the 2026 KEEN Engineering Unleashed

Conference, which promotes entrepreneurial thinking integration in engineering and computing education through evidence-based pedagogy.

2018-2023 Graduate Research Fellowship (GRFP), National Science Foundation

\$138,000 over 3 years

2017 Women-in-Tech Scholar, Adobe Research

\$10,000 to support scholarship and research

Awards & Honors

2026 Faculty Career Champion Award of Excellence

University of Denver

Awarded by student nomination & career council review to a faculty member who has gone above and beyond to support students' career, internship, and post-graduation goals.

2025 Dean's Award, Ritchie School of Engineering and Computer Science

University of Denver

Awarded in first year of assistant professorship for significant contributions to the school's research goals, teaching & mentorship programs, and service to the DU community.

Excellence in Undergraduate Mentoring Award, Student Signature Work Program

University of Denver

2023 Best Paper Award, ACM ICER

2022 Best Paper Award for Diversity & Inclusion Contribution, ACM CSCW

- 2022 **University of Washington Husky 100**
Awarded annually to 100 of UW's 60,000 students who make exceptional contributions to research, teaching, mentorship, and service to the institution. Highest honor available for students.
- 2018 **Outstanding Undergraduate Researcher, Finalist**, Computing Research Association (CRA)
- 2017 **Best Paper Honorable Mention**, ACM CHI
- 2015, 2016 **Drucilla Shepard Smith Academic Excellence Award**, Oregon State University

Invited Talks

- 2026 **Teaching Accessible Computing: A Resource for CS Faculty**
 @ AccessComputing Workshop, ACM SIGCSE Technical Symposium
- 2025 **Toward Critical, Ethical, Creative, and Reflective HCI Education Futures**
 @ ATLAS Colloquium, University of Colorado - Boulder
- Re-Imagining Academia as Collaborative and Community-Centered**
 @ Community Dinner Series, Equity In STEM (E-STEM) Program at U. Denver
- 2024 **Doing HCI Education Research in Your Own Classroom**
 @ Pedagogy Workshop, EduCHI Annual Symposium on HCI Education
- 2021 **Panel: Can Researchers Enhance Diversity and Inclusion?**
 @ International Conference on Software Engineering (ICSE)
Recording available at <https://youtu.be/6LROK9Zqiks>
- 2018 **Research Symposium Welcome Address**
 @ Celebrating Undergraduate Excellence (CUE) Event, Oregon State University

Peer-Reviewed Publications

My publications have been cited over 1,200 times, and I have an h-index of 15 ([Google Scholar](#), June 2026). In Computer Science, conference papers are archival and comparable to journal publications in other fields. Underlined authors below indicate students and junior scholars that I advised and/or mentored.

- 2026 **Visualizing Human-Centered Design: A Replication and Extension Study**
A. Oleson, Y. Kim
 EduCHI Annual Symposium on HCI Education
Replicating and extending a human-centered design (HCD) pedagogical drawing activity originally conducted by Gorichanaz, this study found a "narrowing" effect of computing students' conceptions of HCD over the course of an introductory class, contributing to broader discussions around creativity vs. conformity in HCI and UX learning.
- Starting with DEI and Ethics: A New First-Year Computer Science Introduction**
 S. Leutenegger, S. Hutt, A. Hannum, S. Das, **A. Oleson**, A. Leto, S. Shrestha
 ACM SIGCSE Technical Symposium
Describes a first-year course series implemented at the University of Denver with a focus on diversity, equity, and inclusion, its impacts on students' computer science self-efficacy and attitudes toward ethics, and reflections from instructors on its implementation.

(Under review) **Teaching Accessible Computing: Design & Evaluation of a Community-Sourced Digital Textbook**

A. Oleson, M. Fatima, Y. Kim

ACM SIGCSE Technical Symposium

(Under review) **First Impressions Matter: How CS Syllabi Communicate and Reflect Accessibility Norms**

M. Fatima, A. Oleson

ACM SIGCSE Technical Symposium

(Under review) **From Diary to Discussion: Scaffolding Empathy-Oriented Learning in HCI Education**

J. Fan, A. Oleson, S. Edwards, M. Pérez-Quñones, W. Wang, S. McCrickard

ACM SIGCSE Technical Symposium

(In draft) **AI Policy as a Pragmatic Pathway for Computing Ethics Education**

S. Shrestha, C. Ke, A. Oleson, B. Xie

ACM SIGCSE Technical Symposium

(In draft) **Teaching Critical Interface Design: Challenges, Strategies, and Factors Influencing Success**

A. Oleson et al.

ACM TOCHI: Transactions on Computer-Human Interaction

(In draft) **Computing Students' Conceptions of Roles and Responsibilities in Human-Centered Design**

A. Oleson

ACM CHI: Conference on Human Factors in Computing Systems

(In draft) **Beyond One-Size-Fits-All: Privacy, Security, and Stigma in Telehealth Through the Lens of Marginalized Users**

F. Tazi, S. Saka, Y. Kim, A. Oleson

ACM CHI: Conference on Human Factors in Computing Systems

(In draft) **Measuring Semantic Fidelity in LLM-Based Privacy Policy Interpretation for Older Adults**

S. Saka, A. Oleson, I. Anjum

ACM CHI: Conference on Human Factors in Computing Systems

2025 **Design as Translation, Translation as Design: Toward Critical, Creative, and Ethical Pedagogies**

A. Oleson, L. Taylor

EduCHI Annual Symposium on HCI Education

Explores the intersections of technology design and literary translation studies through a narrative dialogue, with the goal of developing actionable pedagogical insights for helping novices in both fields become more critical, reflective, and ethical practitioners.

Deconstructing Conceptions of Rigor in Computer Science Education

J. Everson, F.M. Kivuva, E. Sherif, A. Oleson, A.J. Ko

ACM TOCE: Transactions on Computing Education

Investigates how K-12 computing instructors, CS undergraduates, and post-secondary computing instructors conceptualize of rigor through a phenomenological approach, finding that definitions of the word widely varied, though it was often used as a vague criteria for worthiness and value in education.

- 2024 **Factors Influencing the Social Help-Seeking Behavior of Introductory Computing Students in a Competitive University Environment**
A.K. Cohen, **A. Oleson**, A.J. Ko
 ACM TOCE: Transactions on Computing Education
Explored how post-secondary CS students sought help on programming assignments, finding both class policies and sociocultural contexts were important in decisions to seek assistance from others.
- 2023 **Teaching Inclusive Design Skills with the CIDER Assumption Elicitation Technique**
A. Oleson, M. Solomon, C. Perdriau, A.J. Ko
 ACM TOCHI: Transactions on Computer-Human Interaction
Contributed a theoretically grounded teaching method to help CS students identify design assumptions that exclude marginalized users and a case study finding lasting impact on their design approaches.
- ★ Best Paper Award
Funds of Knowledge Used by Adolescents of Color in Scaffolded Sensemaking around Algorithmic Fairness
 J. Salac, **A. Oleson**, L. Armstrong, A. Le Meur, A.J. Ko
 ACM ICER: International Computing Education Research Conference
Found that youth in a summer computing camp used a tiered human lens and then technological lens to reason through algorithmic fairness, often invoking facets of their identities and background knowledge.
- 2022 **A Decade of Demographics in Computing Education Research: A Critical Review of Trends in Collection, Reporting, and Use**
A. Oleson*, B. Xie*, J. Salac, J. Everson, F.M. Kivuva, A.J. Ko
 ACM ICER: International Computing Education Research Conference
Found that most computing education research studied adults, left out key details of data gathering, and often used imprecise aggregate terms to illustrate diversity that reified hegemonic identity norms.
 * = co-first authors.
- ★ Best Paper Award for Diversity & Inclusion Contribution
Surfacing Equity Issues in Large Computing Courses with Peer-Ranked, Demographically-Labeled Student Feedback
 B. Xie, **A. Oleson**, J. Everson, A.J. Ko
 PACMHCI: Proceedings of the ACM on Human-Computer Interaction (presented at CSCW)
Developed and evaluated a tool that contextualizes student feedback on learning challenges with anonymized demographic data for teaching teams to identify systemic equity issues within their courses.
- 2021 **On the Role of Design in K-12 Computing Education**
A. Oleson, B. Wortzman, A.J. Ko
 ACM TOCE: Transactions on Computing Education
Disentangled the role of design in K-12 CS education, finding that design ideas are pervasive in curricula and standards, but conflate program space (disciplinary) and problem space (nondisciplinary) design, masking potential challenges to teaching and learning design concepts.
- 2020 **Computing Students' Learning Difficulties in HCI Education**
A. Oleson, M. Solomon, A.J. Ko
 ACM CHI: Conference on Human Factors in Computing Systems
Investigated learning challenges in HCI design classes, finding that students can struggle to differentiate design from engineering, to interpret feedback and scope design problems, and to design for diversity.

Scout: Rapid Exploration of Interface Layout Variations through High-Level Design Constraints

A. Swearngin, C. Wang, **A. Oleson**, A.J. Ko, J. Fogarty

ACM CHI: Conference on Human Factors in Computing Systems

Designed, implemented, and evaluated Scout, a system to support designers' ideation of interface layouts, which helped designers accelerate ideation and produce more diverse design ideas than standard tools.

2018 **Pedagogical Content Knowledge for Teaching Inclusive Design**

A. Oleson, C.J. Mendez, Z. Steine-Hanson, C. Hilderbrand, C. Perdriau, M.M. Burnett, A.J. Ko

ACM ICER: International Computing Education Research Conference

Identified 11 pieces of pedagogical content knowledge for teaching inclusive design in higher education computing courses, including strategies for anticipating and addressing learner resistance, scaffolding perspective-taking, and tailoring instruction to students' prior beliefs.

Semi-Automating (or not) a Socio-Technical Method for Socio-Technical Systems

C.J. Mendez, Z. Steine-Hanson, **A. Oleson**, A. Horvath, C.G. Hill, C. Hilderbrand, A. Sarma, M.M. Burnett

IEEE VL/HCC: Symposium on Visual Languages and Human-Centric Computing

Through a field study and a controlled 92-participant study, discussed the benefits and drawbacks of semi-automating parts of the software interface inspection process from the perspective of cognitive load.

2017 **★ Best Paper Honorable Mention**

Gender-Inclusiveness Personas vs. Stereotyping: Can We Have it Both Ways?

C.G. Hill, M. Haag, **A. Oleson**, C.J. Mendez, N. Marsden, A. Sarma, M.M. Burnett

ACM CHI: Conference on Human Factors in Computing Systems

Found that using multiple profile pictures on one persona may expand product designers' consideration of multiple genders without harming persona engagement or advancing harmful gender stereotypes.

General Principles for a Generalized Idea Garden

W. Jernigan, A. Horvath, M. Lee, M.M. Burnett, T. Culty, S. Kuttal, A. Peters, I. Kwan, F. Bahmani, A.J. Ko, C.J. Mendez, **A. Oleson**

Journal of Visual Languages and Computing

End-user programmers who are not necessarily interested in learning programming can benefit from a just-in-time help system called the Idea Garden, which is built on the presented generalized architecture.

2016 **GenderMag Experiences in the Field: The Whole, the Parts, and the Workload**

C.G. Hill, S. Ernst, **A. Oleson**, A. Horvath, M.M. Burnett

IEEE VL/HCC: Symposium on Visual Languages and Human-Centric Computing

Software practitioners who use the GenderMag method to identify inclusiveness issues in their interfaces engage with personas at a high rate, but may also detour and introduce recording errors during sessions.

Programming, Problem Solving, and Self-Awareness: Effects of Explicit Guidance

D. Loksa, A.J. Ko, W. Jernigan, **A. Oleson**, C.J. Mendez, M.M. Burnett

ACM CHI: Conference on Human Factors in Computing Systems

Teaching novice programmers explicit problem-solving strategies can positively impact their productivity, self-efficacy, independence, and growth mindset development.

Book Contributions & Edited Volumes

- 2024 **CIDER**
Spread in *Universal Methods for Ethical Design: 100 Ways to Become More Ethically Aware, Responsible, and Active in Your Design Work*
Editors: S.S. Chivukula, C. Gray (Quarto Publishing)
A compilation of 100 actionable methods, frameworks, and principles that designers, technologists, and organizations can use to create ethical technology across every design phase.
- 2017 **Toward Theory-Based End-User Software Engineering**
M.M. Burnett, T. Kulesza, **A. Oleson**, S. Ernst, L. Beckwith, J. Cao, W. Jernigan, W. Grigoreanu
Chapter in *New Perspectives in End-User Development*
Editors F. Paternò, V. Wulf (Springer Int'l Publishing)
Highlighted the need for stronger theoretical foundations in end-user software engineering and presented examples of projects that went beyond individual tools to produce general methods and principles.

Digital Books

These books are free, online, accessible, and responsive resources for the computing education community. They are not peer-reviewed in the traditional sense, but instead iteratively improved by feedback from community members in response to the rapidly evolving computing field.

- 2026 **Teaching Accessible Computing**, 2nd ed.
A. Oleson, A.J. Ko, R. Ladner
<https://bookish.press/tac/>
10,000+ readers in 115 countries
Adds three new chapters on integrating accessibility content into Operating Systems, Sustainable Computing, and Capstone/Thesis courses.
- 2024 **Teaching Accessible Computing**, 1st ed.
A. Oleson, A.J. Ko, R. Ladner
<https://bookish.press/tac/>
Offers foundational knowledge about disability and pedagogy to support CS faculty who want to teach about accessibility in their courses, but don't know where to start. The 1st edition of this book contains perspectives from 21 domain experts on how to integrate accessibility into 14 CS courses (intro programming, graphics, data structures, etc.).
- 2022 **Critically Conscious Computing: Methods for Secondary Education**
A.J. Ko, A. Beitlers, B. Wortzman, M. Davidson, **A. Oleson**, M. Kirdani-Ryan, S. Druga, J. Everson
<https://criticallyconsciouscomputing.org/>
A book for secondary educators who want to teach CS from a more critical lens, examining the discipline and its content knowledge from technical, sociotechnical, and sociopolitical stances.
CS and Design (chapter in Critically Conscious Computing)
Authors: A.J. Ko, **A. Oleson**
<https://criticallyconsciouscomputing.org/design>
Describes how design and computing intersect in educational contexts, including the differences between design and engineering mindsets, different pedagogical approaches, and a unit sketch for teaching CS as design-informed engineering.

Articles (Refereed)

- 2020 **It Is Time for More Critical Computing Education**
A.J. Ko, **A. Oleson**, M. Kirdani-Ryan, Y. Register, B. Xie, M. Tari, M.J. Davidson, S. Druga, D. Loksa
ACM CACM: Communications of the ACM
Position article calling for the integration of more critical lenses into computer science education, such as the key ideas that computing has limits, data has limitations, and computer scientists hold responsibility for their creations and design decisions.

Workshops & Symposia

- 2026 **Including Accessibility in Your Computer Science Course**
R.E. Ladner, **A. Oleson**, A.J. Ko
Special Session @ SIGCSE 2026
Proposal for a SIGCSE session focused on bringing together CS educators to discuss challenges and actionable strategies for incorporating accessibility into CS curricula, supported by insights and examples from [Teaching Accessible Computing](#) book authors.
- 2023 **Software Developer Diversity and Inclusion Workshop**
Organizers: D. Ford, E. Murphy-Hill, M. Storey, R. Prikladnicki, Y. Wang
National Institute of Informatics (NII) Shonan Meeting
In the style of Dagstuhl seminars, this invited workshop brought together leading researchers doing diversity and inclusion work in software engineering to identify challenges, propose goals, and gather best practices to share with practitioners. Publication available at <https://shonan.nii.ac.jp/seminars/194/>
- EduCHI 2023: 5th Annual Symposium on HCI Education**
C. Gray, C.M. MacDonald, C. Lallemand, **A. Oleson**, A.R.L. Carter, O. St-Cyr, C. Pitt
EduCHI Annual Symposium on HCI Education @ CHI 2023
Proposal paper for the symposium, formatted like a conference with research talks and teaching demos.
- 2020 **Toward the Development of HCI Pedagogical Content Knowledge**
A. Oleson, A.J. Ko
EduCHI Annual Symposium on HCI Education @ CHI'20
"Unsolved challenge" paper discussing the need for a robust body of HCI pedagogical content knowledge.
- 2019 **The GenderMag-Teach Project**
M.M. Burnett, Z. Steine-Hanson, **A. Oleson**
EduCHI Annual Symposium on HCI Education @ CHI'19
Described how we established and developed an online community of practice for educators teaching gender-inclusive software interface design in their HCI and computing courses.
- 2018 **Gender Biases in Software for Problem-Solving**
M.M. Burnett, A. Sarma, C. Mendez, **A. Oleson**, C. Hilderbrand, Z. Steine-Hanson, A.J. Ko
Designing Technologies to Support Human Problem Solving @ VL/HCC'18
Position paper to call attention to how software can be biased against certain problem-solving styles, especially those favored by women, and how to address these gender-inclusiveness issues.

Patents

- 2018 **Smart Guide to Capture Digital Images that Align with a Target Image Model**
A. Oleson, R. Mech, J. Echevarria, C. Lu

Filed as US15/897,951; published as US20190253614A1 and US10574881B2

Describes an interface model for analyzing a mobile device's camera feed as a picture is being taken and guiding a user to achieve a higher-quality selfie or portrait in real time.

Dissertation

2023 **Integrating Inclusive Design and Computing Education**

A. Oleson

For a Ph.D. in Information Science at the University of Washington

[\[ProQuest Link\]](#)

Doctoral Consortia

2022 **CIDER: A Method to Teach Practical Critical Software Design Skills**

A. Oleson

SIGCSE DC @ ACM ICER Conference on International Computing Education Research

Position article calling for the integration of more critical lenses into computer science education, such as the key ideas that computing has limits, data has limitations, and computer scientists hold responsibility for their creations and design decisions.

2022 **Supporting Critical Software Design Decision-Making in HCI Education**

A. Oleson

DUB DC @ University of Washington

Describes past and proposed research on how to help computing students think more critically about the impacts of their design decisions so that they can create more usable, accessible, and inclusive software.

Teaching Experience

Assistant **Design Justice**, DU COMP 3701 (Spring 2026)

Professor *New course, original curriculum design. Seminar. Students read about, discuss, and reflect upon what “justice” means in the context of today’s technological landscape from different theoretical perspectives (e.g. disability justice, anti-racist design), articulating and refining their values and commitments as tomorrow’s technology practitioners.*

User Experience Design Methods, DU COMP 3705 (Winter 2026)

New course, original curriculum design. Flipped classroom. Introduces students to the basic methods needed to succeed in UX design careers, such as problem framing, ideation methods, low- and mid-fidelity prototyping, design evaluation methods, and synthesis of user research. Students deeply explore a design problem of interest from multiple perspectives and present a design specification at the end of the term.

Human-Computer Interaction, DU COMP 3/4100 (Fall 2025, Winter 2025)

Redesigned an introductory HCI/UX design course (elective for graduates & undergraduates) to integrate active learning and discussion activities in every class session, based on effective HCI education research. Students learn basic HCI principles and design methods through digital prototyping, then build collaboration skills through a final design team project to address a structural inequity around campus.

Introduction to Computer Science, DU COMP 1201 (Fall 2024)

Designed and implemented (in collaboration with colleagues) an introductory CS course (required for majors/minors, open to all) on algorithmic thinking, career readiness, and ethical perspectives on technology. Students engage in activities and discussions to build CS foundations and identities.

Instructor of Record	User-Centered Design Methods , UW INFO 360 (Spring 2020) <i>Adapted course materials to support more equitable participation during remote learning; Created original assignments to help students recognize and respond to exclusionary design biases.</i>
Teaching Assistant	Cooperative Software Development , UW INFO 442 (Fall 2020) <i>Helped adapt course content to equitable remote instruction formats, focusing on needs of the most disadvantaged student groups first, and implementing structures for instructor and peer support.</i>
	User-Centered Design Methods , UW INFO 360 (Winter 2019) <i>Created formative assessment questions to determine gaps in students' understandings and adapt instruction accordingly; Prepared and delivered lesson on interface evaluation methods.</i>
Guest Instructor	Computing Education Research Seminar , UW CSE 599 (Winter 2024) <i>Led discussion on the idea of "CS For All" and how to equitably expand CS education access.</i>
	Design Methods , UW INFO 360 (Winter 2019) <i>Presented an overview of the principles of various empirical and analytical user interface evaluation methods and led a series of active learning activities for students to engage with each method.</i>

Students Supervised

I have meaningfully mentored 23 students (9 grad, 14 undergrad, 2 high school), including several women, non-binary, and transgender people, many Students of Color, several international students, a few from rural areas, and several students from economically disadvantaged backgrounds, with varied (dis)abilities and prior exposure to academia. 15 of these students co-authored academic papers with me. At least 11 of the undergraduates and high schoolers chose to pursue graduate education after working with me.

Professional Service

Steering Committee	EduCHI Symposium on HCI Education (2023-present) <i>Providing oversight and guidance to the organizing committee during the transition to a standalone conference in 2024; Liaison to ACM TOCE journal and co-director for early-career researcher workshops.</i>
Partner	AccessComputing , University of Denver (2024-present) <i>Acting as DU's liaison to the AccessComputing program to support participation and access for computing students with disabilities; Connecting DU faculty and staff to accessible computing resources.</i>
Associate Editor	ACM Transactions on Computing Education (2024-2025) <i>Expertise areas: Human-computer interaction (HCI) education and computing design education research.</i>
Technical Program Chair	EduCHI Symposium on HCI Education (2023, 2025) <i>Oversaw the paper review process for the symposium's research and teaching-related submissions and worked with program committee to make final acceptance decisions; Collaborated with the General Chairs to create and promote community-building workshops prior to the symposium.</i>
Departmental Service	• Tenure-Track Hiring Committee, 2025-2026 • Graduate Admissions Restructuring Committee, 2025-2026 • Graduate Curriculum Committee, 2024-2025 • Undergraduate Honors, Awards, and Distinction Committee, 2024-2025
School Service	• Academic Appeal Committee (ad hoc), 2025

- Reviewer **MIT Press**; ACM **CHI** Conf. on Computer-Human Interaction; ACM **CSCW** Conf. on Computer-Supported Collaborative Work; ACM **TOCE** (Trans. on Computing Education); ACM **EduCHI** Conf. on HCI Education; ACM **TOSEM** (Trans. on Software Engineering Methodology); **Int'l Journal of Child-Computer Interaction**; ACM **SIGCSE** Technical Symposium
- Discussant **RESPECT Conference Doctoral Consortium** (2025)
- Discussant **Humanization of Computing and Engineering Education Doctoral Consortium** (2025)
- Organizer **HCI Pedagogy Workshop**, EduCHI Symposium (2024)
Planned and led the inaugural half-day workshop on HCI pedagogy and pedagogical research for doctoral students, postdocs, and 1st year faculty, which became a recurring event due to its success.
- Mentor **Trans/Queer in HCI Mentorship Program** (2022-2023)
Providing professional and personal support to students in the Human-Computer Interaction field who are part of the LGBTQIA2S+ community.
- Chair **UW iSchool Doctoral Student Association** (2021-2022)
Organized and presided at DSA meetings and town halls; Led DSA outreach at new student orientation; Advocated for student perspectives to faculty, staff, and administrators.
- Officer **Secretary**, UW iSchool Doctoral Student Association (2020-2021)
Organized and disseminated information internally to doctoral students, including meeting minutes; oversaw internal DSA officer elections.
Communications & Outreach, UW iSchool Doctoral Student Association (2019-2020)
Administration of internal \& external communications: mailing lists, social media accounts, reports on DSA activity, other outreach channels as requested.
- Student ACM **CHI** Conference (2019, 20, 22, 23), ACM **SIGCSE** Technical Symposium (2019)
- Volunteer ★ *CHI 2019: Received internal SV award "for going above and beyond" typical SV duties to ensure conference participants' accessibility needs were properly met during paper sessions.*
- Student **UW DUB Seminar** (2019-2020)
- Coordinator *DUB (Design, Use, Build) is an interdisciplinary community at the UW focused on HCI and Design which coordinates weekly invited talks with internal and external speakers.*
- Peer Mentor **OSU STEM Leaders Program** (2015-2018)
Helped 5 freshmen from underrepresented backgrounds in STEM transition to and succeed in new college environments as they completed original research projects with faculty mentors.